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| Revision History |      |            |              |                                 |
|------------------|------|------------|--------------|---------------------------------|
| Marketing PN     | Rev. | Date       | By           | Description                     |
| X-NAVQ95-MAIN    | X1   | 2026-05-07 | JvE (CTO SI) | Initial Schematic Development   |
| X-NAVQ95-MAIN    | A    | 2026-06-11 | JvE (CTO SI) | Schematic release               |
| NAVQ95-MAIN      | A1   | 2026-09-10 | JvE (CTO SI) | Schematic release to production |



#### NOTES:

1. Interrupted lines coded with the same letter or letter combinations are electrically connected.
2. Device type number is for reference only. The number varies with the manufacturer.
3. Greyed out components indicate Do Not Populate (DNP) during assembly.
4. Special signal usage:
  - \_B denotes an active-low signal.
  - \_P and \_N denote a differential signal.
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.
6. All SILK display properties will be in this color
7. All schematic annotations will be in this color

|                                                                                                                                                                                                                  |                                      |                                                                                       |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------|-----------|
|                                                                                                                             |                                      | <b>System Innovation</b><br>High Tech Campus 90<br>5656 AG Eindhoven, The Netherlands |           |
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|                                                                                                                             |                                      | Classification: <b>Public</b>                                                         |           |
| Designer:<br>Jari van Ewijk                                                                                                                                                                                      | Drawing Title:<br><b>NAVQ95-MAIN</b> |                                                                                       |           |
| Drawn by:<br>Jari van Ewijk                                                                                                                                                                                      | Page Title:<br><b>Cover</b>          |                                                                                       |           |
| Approved:<br>Cristin Onofrei                                                                                                                                                                                     | Size<br>C                            | Document Number<br>SCH-97010 / PDF: SPF-97010                                         | Rev<br>A1 |
| Date: Thursday, September 17, 2026   Sheet 1 of 26                                                                                                                                                               |                                      |                                                                                       |           |

## DNP Components

| REF DES                                                             | ASSY OPT | PAGE NAME               |
|---------------------------------------------------------------------|----------|-------------------------|
| R124,R126,R269                                                      | DNP      | 05 - HV Buck Converter  |
| C23,C29,C92,R65,R68,R73,R74,<br>R145,R152,R156,R229,R230,R257       | DNP      | 06 - LV PMICs           |
| R112,R123,R137,R138,R139,R254                                       | DNP      | 09 - MPU I/O            |
| C181,D14,R250                                                       | DNP      | 10 - MPU Misc.          |
| R292,R295,R296,R297,R298                                            | DNP      | 11 - eMMC Flash         |
| R141                                                                | DNP      | 12 - Octal SPI Flash    |
| C81,R288,R290                                                       | DNP      | 13 - MicroSD Card       |
| R52,R56,R60,R75,R78                                                 | DNP      | 17 - RTC & Clock Buffer |
| R378,R379                                                           | DNP      | 18 - Sensors            |
| C427,R158,R159,R162,R164,<br>R165,R166,R169,R361,R362,<br>R364,R365 | DNP      | 19 - Tri-Radio          |
| R315,R317                                                           | DNP      | 20 - Gigabit Ethernet   |
| R212                                                                | DNP      | 21 - M.2 Key M          |
| R113,R114,R115,R266,R267,<br>R268,R289                              | DNP      | 22 - M.2 Key B          |
| R103,R176,R177,R178,R179,<br>R180,R181                              | DNP      | 23 - I/O Expanders      |
| R185                                                                | DNP      | 24 - Secure Element     |

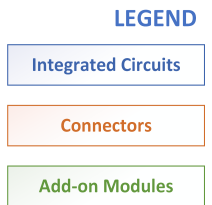
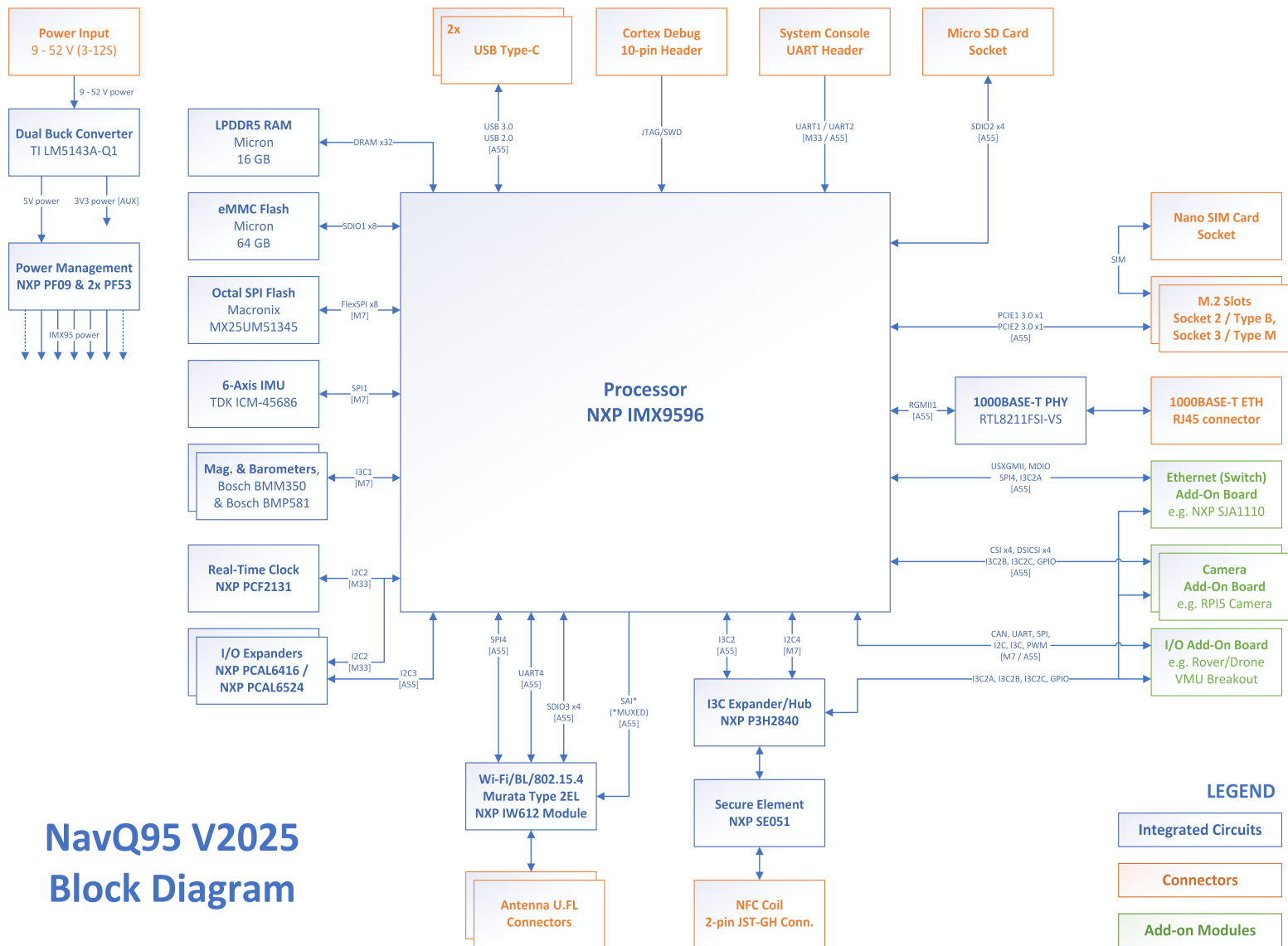
## Switch Default Positions

| REF DES | SWITCH  | PAGE NAME                |
|---------|---------|--------------------------|
| SW2     | 2-4-5-8 | 14 - Buttons & Boot Mode |



|                                    |                                            |        |
|------------------------------------|--------------------------------------------|--------|
| Classification: Public             |                                            |        |
| Drawing Title: NAVQ95-MAIN         |                                            |        |
| Page Title: Tables                 |                                            |        |
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# NavQ95 V2025 Block Diagram



Molex Micro-Fit 2-pin power input  
3S-12S BAT IN (6S recommended)  
9-52V / 8A

LM5143 Buck Converter

PF09 / PF5301 / PF5302  
VAON = 1.8V/10mA  
VDD\_SOC, PF5302 = 0.61-0.955V/15A  
VDD\_ARM, PF5301 = 0.61-1.05V/8A  
SW2 = 0.8375/2.5A  
SW3 = 1.8/2.5A  
SW5 = 1.05-1.1V/2.5A  
SW4 = 0.5-0.6V/2.5A  
SW1 = 3.3V/3.5A  
LDO2 = 1.8-3.3V/200mA  
LDO1 = 3.3V/500mA  
LDO3 = 0.65V/200mA  
RSTB

IMX95  
NVCC\_BB5M\_1V8 (2mA req.)  
VDD\_SOC (14A req.)  
VDD\_ARM (6.5A req.)  
VDD\_ANA\_0P8 (1.5A req.)  
VDD\_ANAx\_1P8 (300mA req.)  
VDD\_GPIO\_1P8 (400mA req.)  
VDD2\_DDR (50mA req.)  
VDDQ2\_DDR (250mA req.)  
VDD\_GPIO\_3P3 (400mA req.)  
NVCC\_SD2 (200mA req.)  
SD\_CARD (400mA req.)  
POR\_B

LPDDR5  
VDD1 = 1.8V/50mA  
VDD2 = 1.05V/1.2 A  
VDDQ = 0.5V/300mA

eMMC  
VCC = 3.3V/200mA  
VCCQ = 1.8V/250mA

MicroSD  
VDD = 3.3V  
VIO = 1.8-3.3V

OctalSPI Flash  
MX25UM51345GXD100  
VCC = 1.8V/80mA

RTC  
VDD = 3.3V/100nA  
BACKUP BATTERY

IW612  
AVDD18 = 1.8V/1A  
AVDD33 = 3.3V/500mA

RTL8211F5  
VDD = 3.3V/1A

156.25MHz  
ETH REFCLK  
e.g. 511DBA156M250BAG  
VDD = 3.3V/50mA

100MHz  
PCIe Clock Buffer  
e.g. S153204-A01AGM  
VDD = 1.8V/50mA

Sensors  
3.3V/50mA  
1.8V/50mA

I/O Expanders  
3.3V/250mA  
1.8V/250mA

RPI5-Compatible RTC Backup Battery  
JST-SH 2-pin connector

USB PD  
VDD\_SYS\_SV / 4.5A

I/O ADD-ON  
VDD\_SYS\_SV / 1A  
VDD\_SYS\_3V3 / 2A

ETH ADD-ON  
VDD\_SYS\_SV / 1A  
VDD\_SYS\_3V3 / 2A

CAM ADD-ON  
VDD\_SYS\_SV / 1A  
VDD\_SYS\_3V3 / 2A

M.2 Key B  
VDD\_SYS\_3V3 / 2.5A  
(5A peak for 100us)

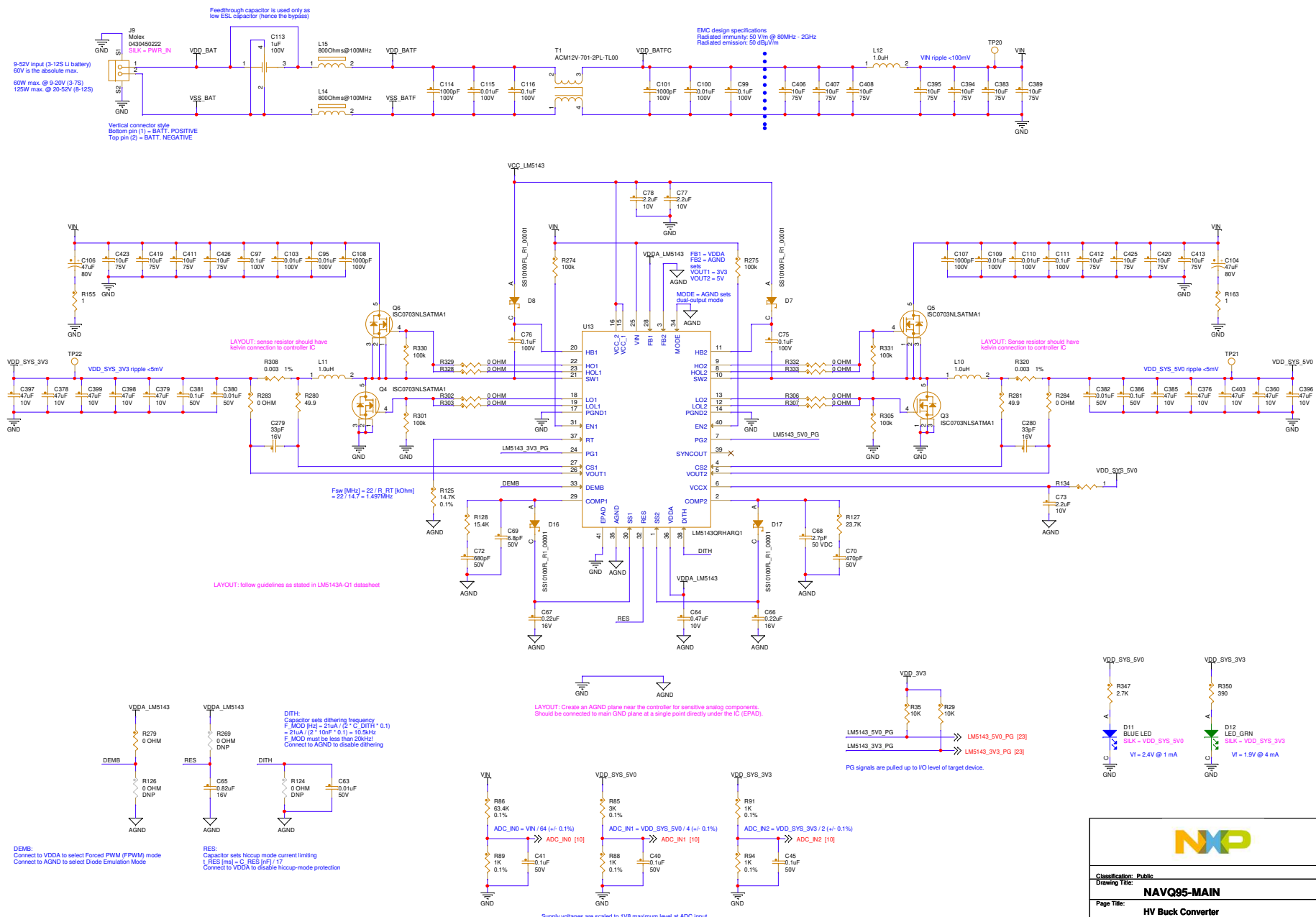
M.2 Key M  
VDD\_SYS\_3V3 / 3.5A  
(7A peak for 100us)

## NavQ95 V2025 Power Tree

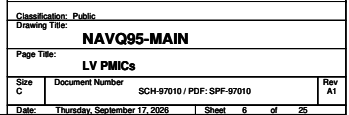


|                            |                              |                            |        |
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| Drawing Title: NAVQ95-MAIN |                              |                            |        |
| Page Title: Power Tree     |                              |                            |        |
| Size C                     | Document Number              | SCH-97010 / PDF: SPF-97010 | Rev A1 |
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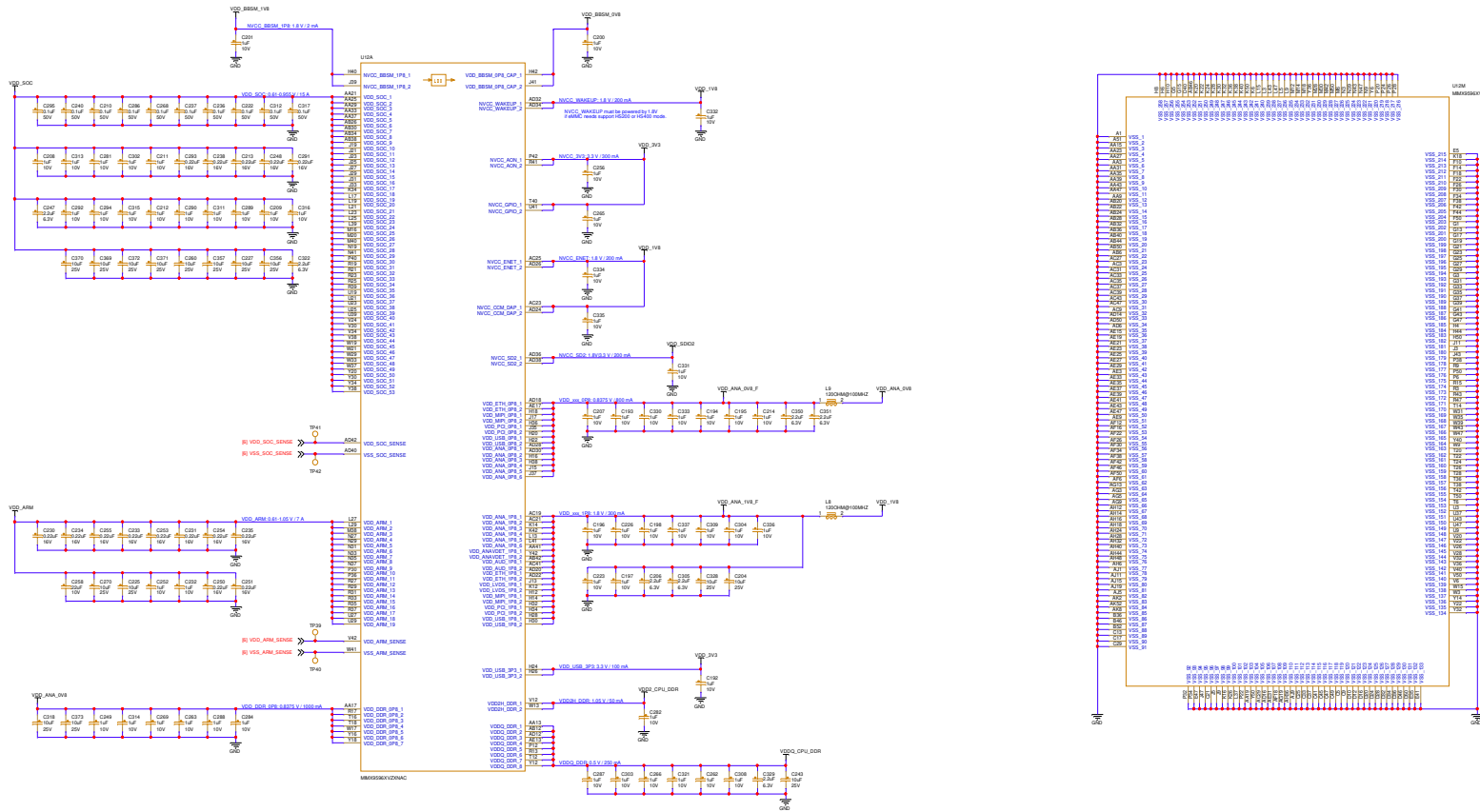
# HV Buck Converter



|                                                                                       |                                               |       |   |           |
|---------------------------------------------------------------------------------------|-----------------------------------------------|-------|---|-----------|
|  |                                               |       |   |           |
| Classification: Public                                                                |                                               |       |   |           |
| Drawing Title:                                                                        |                                               |       |   |           |
| <h1>NAVQ95-MAIN</h1>                                                                  |                                               |       |   |           |
| Package Title:                                                                        |                                               |       |   |           |
| <h2>HV Buck Converter</h2>                                                            |                                               |       |   |           |
| Size<br>C                                                                             | Document Number<br>SCH-97010 / PDF: SPF-97010 |       |   | Rev<br>A1 |
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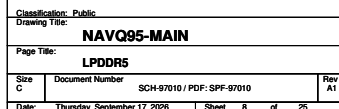
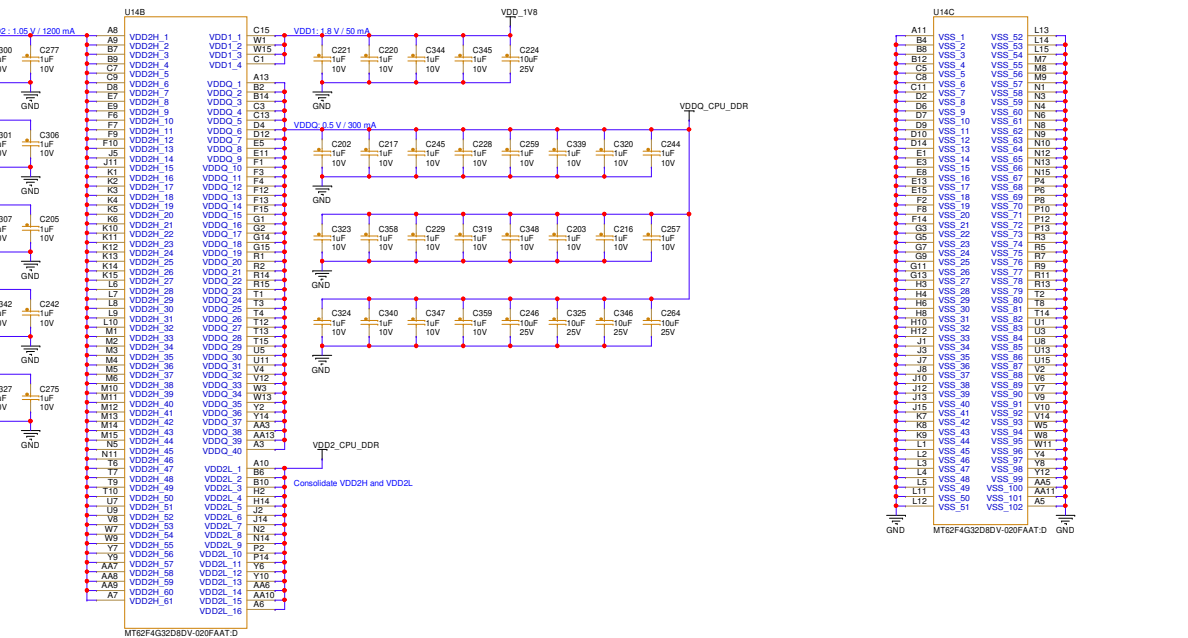


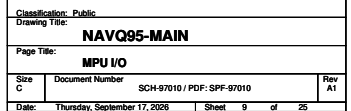
## MPU Power



Note: net names are re-arranged (swizzled) to allow for optimized routing. CA net name follows memory device name. DQ net name follows CPU signal name.

DDR: 80 ohm



[illegible]

**Caution:**

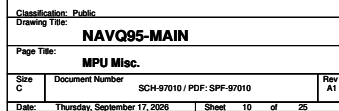
# At the first time of BBSM\_1P8 ramping up,  
 -PMIC\_ON\_REG / PMIC\_STBY\_REG are HIGH+Z;  
 about 1ms delay  
 -PMIC\_ON\_REG strong output HIGH;  
 -PMIC\_STBY\_REG strong output LOW

Freq: 32.768 kHz  
 Freq TOL: +/- 20 ppm  
 CL: 7 pF  
 DL: Max: 0.5 uW  
 ESR: 80 kohm  
 MPN: NX3215SA 32.768kHz-EX300A-MU01567

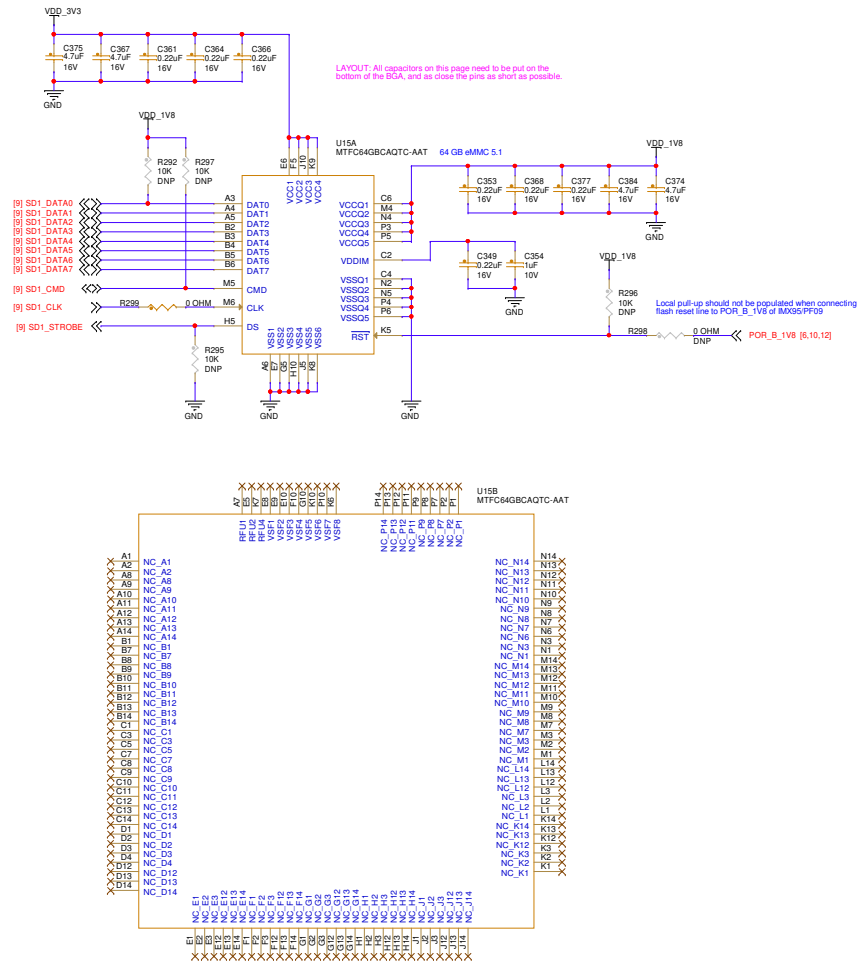
Freq: 24 MHz  
 Freq TOL: +/- 30 ppm  
 CL: 8 pF  
 DL: Max: 300 uW  
 ESR: 100 ohm  
 MPN: XHC024M000F3A13R0

Connector J2 follows DS-009 Pinhawk connector standard  
 for UART / TELEM pin 2 = UART\_TXD / 3 = RXD / 4 = CTS / 5 = RTS  
 UART2\_RXD = UART1\_CTS, B ALT MODE  
 UART2\_TXD = UART1\_RTS, B ALT MODE

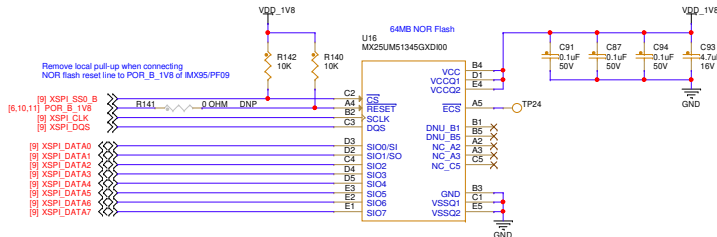
An external debugger may only pull RST\_B low (sink current) and should not be able to drive the RST\_B signal high.  
 To avoid backfeeding VDD\_1V8 an additional diode is added in series with the local pull-up resistor.



# eMMC Flash

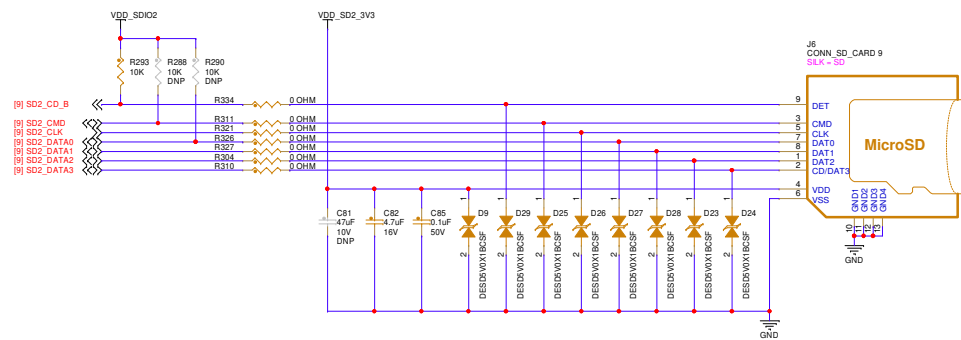


## Octal SPI Flash



|                                       |                                               |       |          |
|---------------------------------------|-----------------------------------------------|-------|----------|
| Classification: Public                |                                               |       |          |
| Drawing Title:<br><b>NAVQ95-MAIN</b>  |                                               |       |          |
| Page Title:<br><b>Octal SPI Flash</b> |                                               |       |          |
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# MicroSD Card



|                                    |                                            |        |  |
|------------------------------------|--------------------------------------------|--------|--|
| Classification: Public             |                                            |        |  |
| Drawing Title: NAVQ95-MAIN         |                                            |        |  |
| Page Title: MicroSD Card           |                                            |        |  |
| Size C                             | Document Number SCH-97010 / PDF: SPF-97010 | Rev A1 |  |
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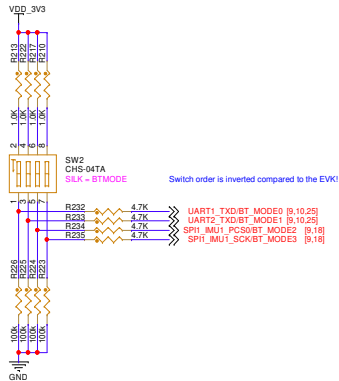
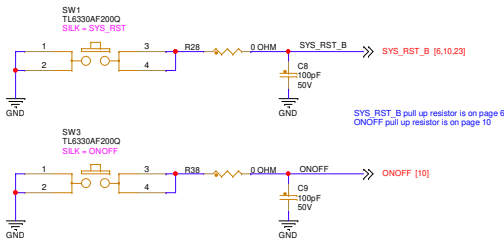
# Buttons & Boot Mode

## SYS RESET

Short Press: Reset

## ON/OFF

Long Press 5 seconds: Turn OFF  
Short Press: Turn ON



## i.MX95 Boot Modes

| BOOT_MODE[3:0] | SW2-1 | SW2-2 | SW2-3 | SW2-4 | BOOT DEVICE                     |
|----------------|-------|-------|-------|-------|---------------------------------|
| 1001           | ON    | OFF   | OFF   | ON    | Serial Downloader on USB1 (J13) |
| 1010           | OFF   | ON    | OFF   | ON    | Boot from eMMC Flash (U15)      |
| 1011           | ON    | ON    | OFF   | ON    | Boot from MicroSD Card (J6)     |
| 1100           | OFF   | OFF   | ON    | ON    | Boot from Octal SPI Flash (U16) |

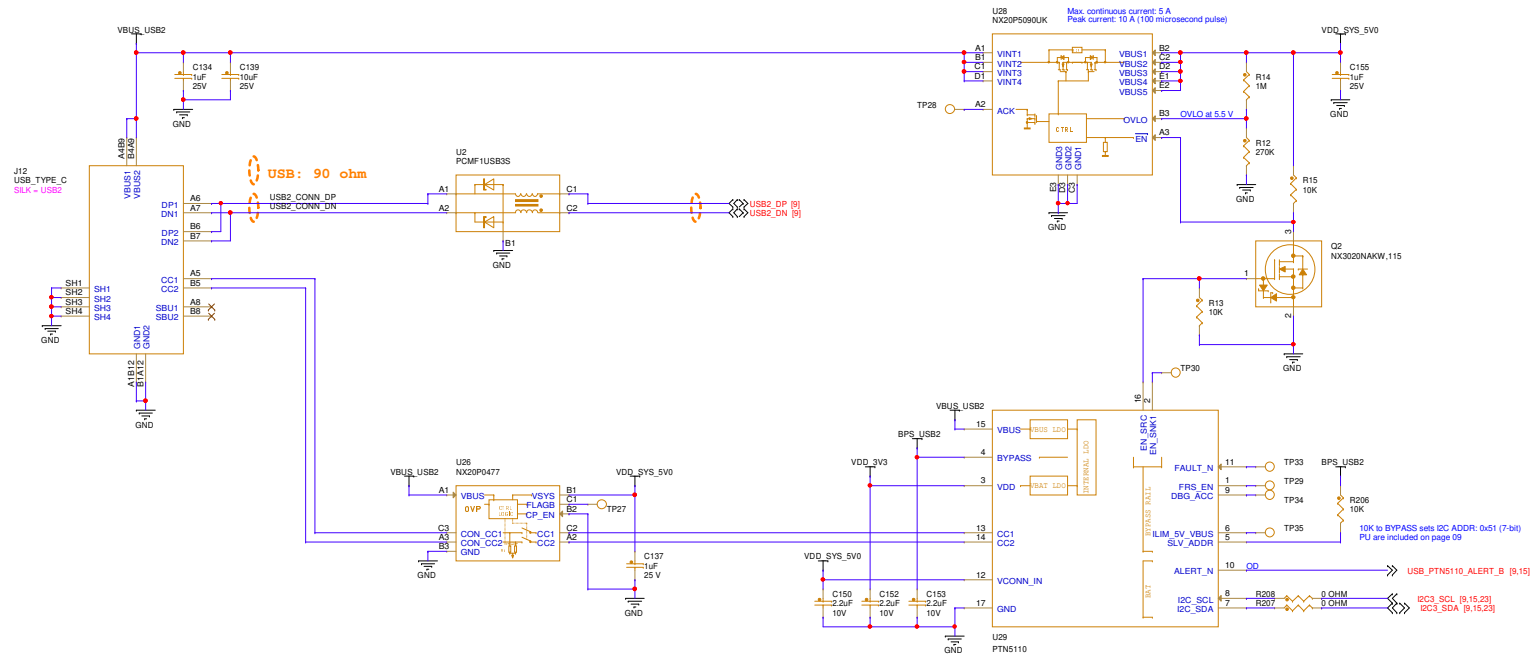
Switch order is inverted compared to the EVK!  
See MX95M for more details about different boot modes.



|                                 |                              |                            |        |
|---------------------------------|------------------------------|----------------------------|--------|
| Classification: Public          |                              |                            |        |
| Drawing Title: NAVQ95-MAIN      |                              |                            |        |
| Page Title: Buttons & Boot Mode |                              |                            |        |
| Size C                          | Document Number              | SCH-97010 / PDF: SPF-97010 | Rev A1 |
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# USB2 2.0 Type C

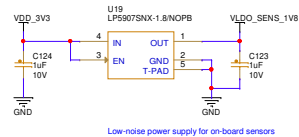


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| Page Title: USB2 2.0 Type C        |                                            |        |  |
| Size C                             | Document Number SCH-97010 / PDF: SPF-97010 | Rev A1 |  |
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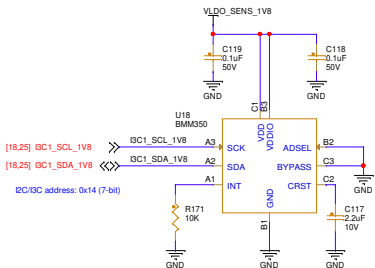


# Sensors

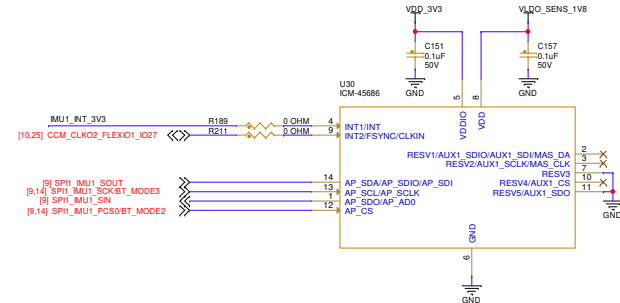
## Low-Noise Regulator



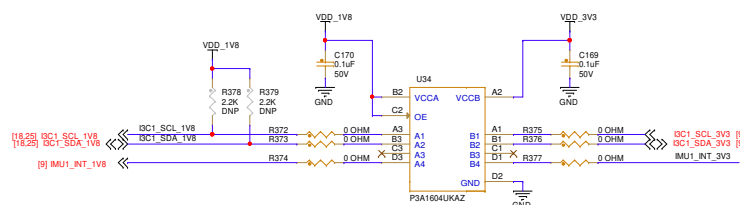
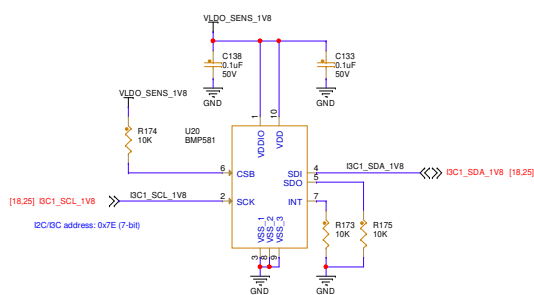
## Magnetometer



## 6-Axis Accelerometer & Gyroscope



## Barometer



|                                    |                                            |        |  |
|------------------------------------|--------------------------------------------|--------|--|
| Classification: Public             |                                            |        |  |
| Drawing Title: NAVQ95-MAIN         |                                            |        |  |
| Page Title: Sensors                |                                            |        |  |
| Size C                             | Document Number SCH-97010 / PDF: SPF-97010 | Rev A1 |  |
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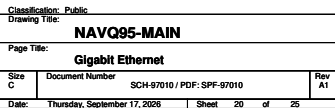
## Wi-Fi 6 / Bluetooth 5.4 / IEEE 802.15.4



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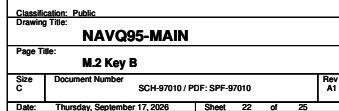
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# Gigabit Ethernet

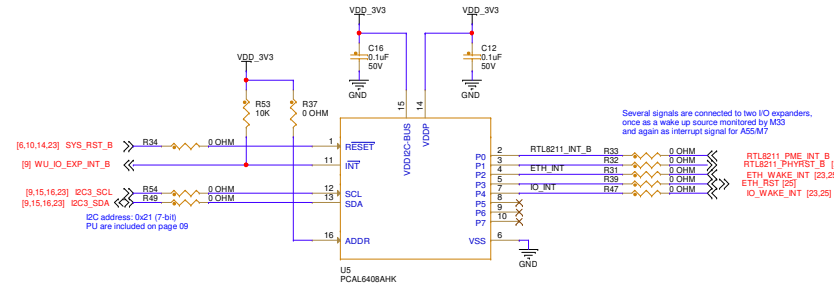
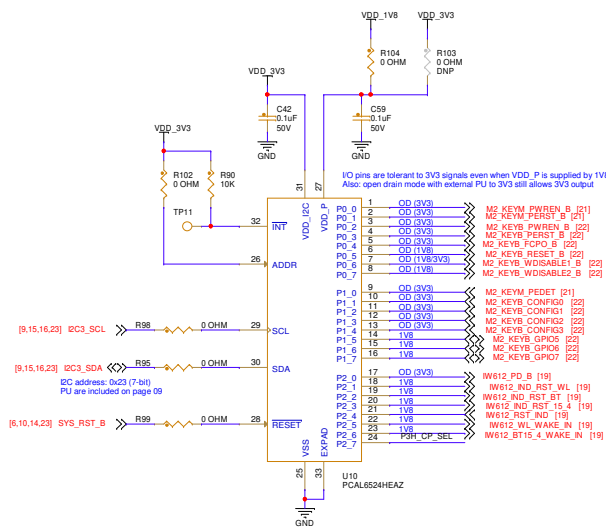
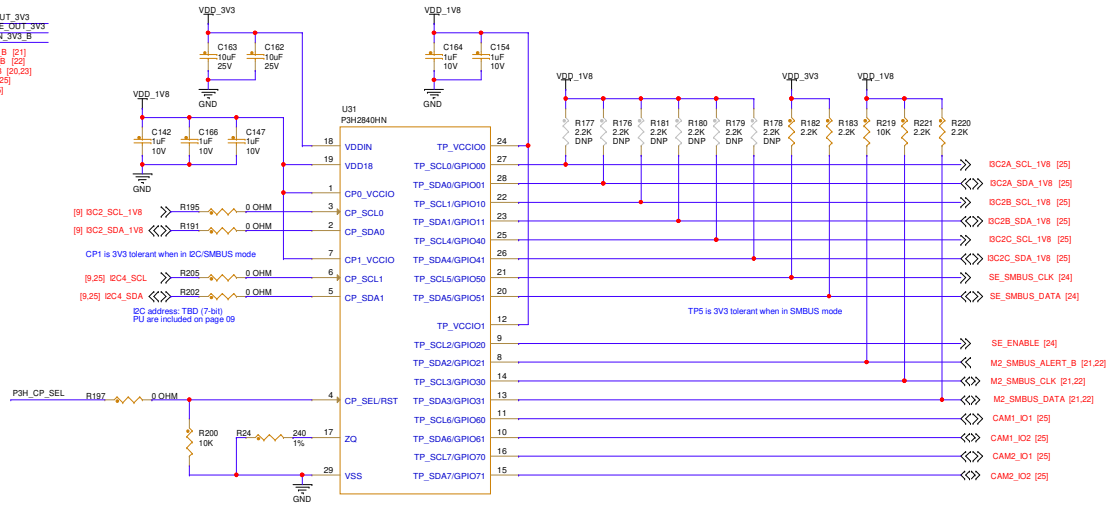
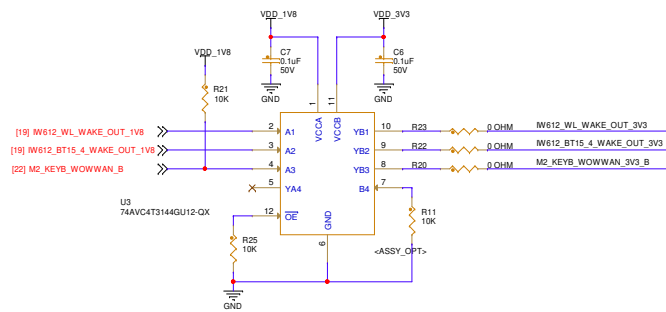
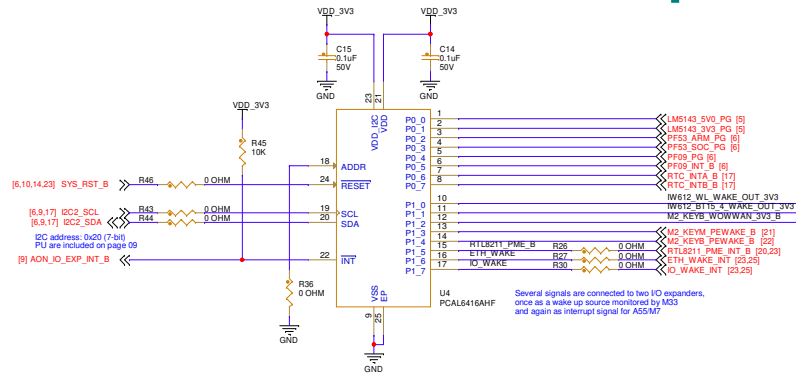




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| P |
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# I/O Expanders



Classification: Public  
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Drawing Title: **N**

NAVQ95-MAIN

Page Title:

## I/O Expanders

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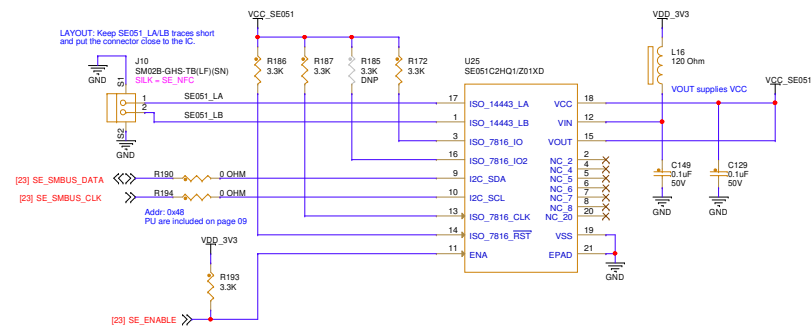
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Thursday, September 10, 2008 10:00 AM

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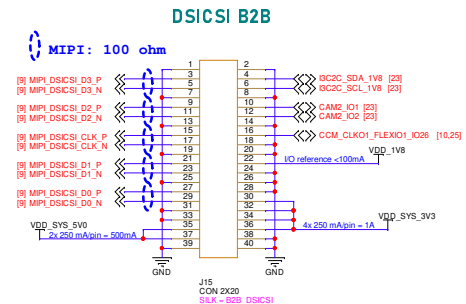
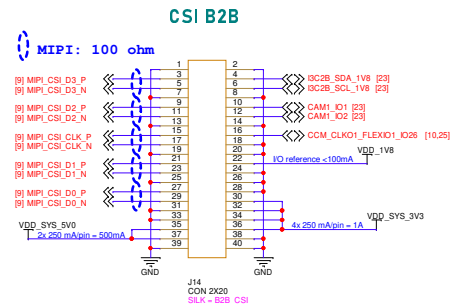
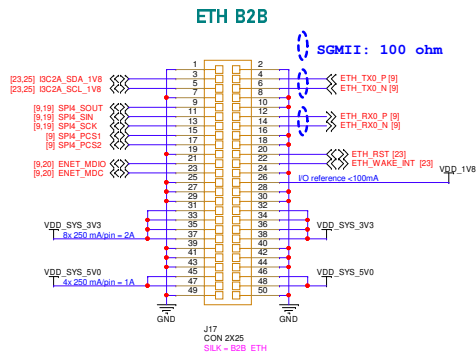
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# Secure Element



|                                    |                                            |        |  |
|------------------------------------|--------------------------------------------|--------|--|
| Classification: Public             |                                            |        |  |
| Drawing Title: NAVQ95-MAIN         |                                            |        |  |
| Page Title: Secure Element         |                                            |        |  |
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# Add-On B2B Interfaces



## I/O B2B

